Website Funnel Analysis

Final Project Report

Project: Website Funnel Analysis
Technology: SQL and Microsoft Power BI
Dataset: client_site_dataset.csv

---

1. Project Overview

The Website Funnel Analysis project focuses on understanding customer behaviour on an e-commerce website. The objective is to analyze how users move through different shopping stages, identify where customers leave the purchasing journey, and understand revenue performance across marketing channels, regions, devices, and product categories.

SQL was used for data exploration and business analysis, while Power BI was used to create an interactive dashboard for visualizing the findings.

2. Project Objectives

- Analyze user behaviour throughout the shopping journey.
- Measure customer conversion at each funnel stage.
- Identify major customer drop-off points.
- Compare revenue across marketing channels and regions.
- Analyze device and product category performance.
- Develop actionable business recommendations.

3. Tools and Technologies

Tool	Purpose
SQL	Data exploration and analysis
Microsoft Power BI	Dashboard development and visualization
DAX	Measures and calculated columns
Power Query	Data cleaning and transformation
CSV	Source dataset

4. Dataset Overview

The project uses client_site_dataset.csv.

The dataset contains **21,409 records and 10 columns**.

Important columns include:

- User ID
- Session ID
- Event Time
- Event
- Device
- Region
- Channel
- Product Category
- Revenue
- Bonus Flag

The dataset contains 10,000 unique users and 10,000 unique sessions.

5. Methodology

Step 1: Data Exploration

- Examined the dataset structure and column types.
- Checked record counts and unique users.
- Identified event types and their frequencies.
- Examined missing values and data quality.

Step 2: SQL Analysis

SQL queries were used to calculate:

- Total users and events.
- Funnel stage counts.
- Conversion and drop-off percentages.
- Revenue by channel, region, device, and product category.
- Top revenue-generating customers.

- Device conversion performance.

Step 3: Power BI Development

- Imported and cleaned the dataset.
- Created DAX measures.
- Developed an interactive dashboard.
- Added KPI cards, funnel chart, revenue charts, device distribution, and drop-off analysis.
- Added slicers for Region, Device, Channel, and Product Category.

6. Key Performance Indicators

KPI	Result
Total Users	10,000
Total Revenue	277,323.06
Total Events	21,409
Total Purchases	1,004
Overall Purchase Conversion	10.04%

7. Funnel Analysis

The actual event journey available in the dataset is:

Funnel Stage	Users
Browse	10,000
Add to Cart	6,949
Checkout	3,456
Purchase	1,004

Funnel Findings

- Browse to Add to Cart conversion: 69.49%.
- Add to Cart to Checkout conversion: 49.73%.
- Checkout to Purchase conversion: 29.05%.
- Overall Browse to Purchase conversion: 10.04%.

The largest absolute drop-off occurs between Add to Cart and Checkout, with 3,493 users lost.

The highest percentage drop-off occurs between Checkout and Purchase, at 70.95%.

8. Revenue Analysis

Revenue by Channel

Channel	Revenue
Google Ads	73,862.32
Email	69,126.46
Social Media	68,361.24
Organic	65,973.04

Google Ads generated the highest revenue among the available channels.

Revenue by Region

Region	Revenue
South	77,421.45
North	68,645.13
East	66,116.01
West	65,140.47

South generated the highest regional revenue.

Revenue by Device

Device	Revenue
Desktop	98,471.83
Tablet	94,620.13
Mobile	84,231.10

Desktop generated the highest revenue.

Revenue by Product Category

Product Category	Revenue
Electronics	62,938.46
Fashion	58,075.22
Sports	55,856.01
Beauty	50,353.36
Home	50,100.01

Electronics was the highest-revenue product category.

9. Device Conversion Analysis

Device	Browse Users	Purchasers	Conversion Rate
Desktop	3,366	356	10.58%
Tablet	3,371	339	10.06%
Mobile	3,263	309	9.47%

Desktop achieved the highest purchase conversion rate, while Mobile had the lowest.

10. Business Insights

1. Only 10.04% of browsing users completed a purchase.
2. The largest percentage drop-off occurred during the final checkout-to-purchase stage.
3. Google Ads generated the highest channel revenue.
4. South was the highest-revenue region.
5. Desktop users generated the highest revenue and conversion rate.
6. Electronics generated the highest revenue among product categories.

11. Business Recommendations

- Simplify the checkout process to reduce customer abandonment.
- Investigate payment failures and unexpected checkout costs.
- Evaluate marketing campaigns using both revenue and conversion performance.
- Study successful sales strategies in the South region.
- Improve mobile shopping navigation and checkout usability.
- Explore ways to increase sales in the Home product category.

12. Conclusion

The Website Funnel Analysis project demonstrates how SQL and Power BI can be used together to understand customer behaviour and business performance.

The analysis identified important customer drop-off points, high-performing marketing channels, regional revenue differences, and device-level conversion patterns.

The findings can help businesses improve their purchasing journey, optimize marketing activities, and make data-driven decisions to improve customer conversion and revenue.

13. Project Limitations

- The project description originally mentioned five funnel stages, but the supplied dataset supports four actual event types: Browse, Add to Cart, Checkout, and Purchase.
- Event Time is stored as text and does not provide a verified calendar date for reliable revenue trend analysis.
- Revenue currency is not specified in the supplied dataset.
- Marketing profitability and return on advertising spend cannot be established without campaign cost data.